Introduction to Economics

Class 9 — Price controls; international trade and tariffs

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Universitat de Barcelona · Group A6

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slide 1 of 28

Slide 1 Title

Block 3 continues. Monday measured what a market is worth and put a tax on it; today the lost triangle gets a formula and is then used to judge two more things a government does: setting the price itself, with a rent cap and a minimum wage, and taxing goods that cross the border. The thread is a real rent cap in Catalonia: what the picture says it should do, and what happened next.

Quiz 6

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top.

A question marked + can have more than one correct letter: circle all of them.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

quiz page, not numbered

Quiz Start of class, on paper

A short quiz on paper opened the class: circle the letter of your answer in the block at the top of the sheet, write your name on the block, tear the strip off along the dashed line and hand in only the strip.

Plan for today

10:00	Quiz 6
10:11	Warm-up
10:15	Catalonia caps rents: what will happen?
10:18	The lost triangle, computed
10:29	In pairs: taxes on menús
10:34	A rent cap: a shortage, now and over the years
10:43	Catalonia: what the data say
10:46	At the board: a cap on allotment rents
10:50	Break · Problem set 1 on the front desk
11:00	A price floor: the minimum wage
11:04	Open borders: who wins, who loses
11:13	In pairs: cava at home and abroad
11:19	A tariff: who gains, what is lost
11:27	European Union (EU) car duties; why limit trade; in pairs
11:39	At the board: a tax on driving lessons
11:44	In one line; before Monday 26 October

Slide 2 Plan for today

The class at a glance, and one tool for all of it. Monday's lost triangle first gets a formula and three rules; the same triangle then measures what a rent cap and a minimum wage cost, and after the break what opening a border gains and what a tariff takes back. The hook is Catalonia's rent cap: the picture predicts what it should do, and the end of the first half says what the data show.

Two halves of fifty minutes with a ten-minute break between them; Problem set 1 is handed in at the break.

Warm-up: which elasticity, what number, what word?

4 % fewer tubs of margarine sold. Two different causes:

1. Its price rose 10 %; nothing else changed
   price elasticity of demand: $|-4\,\% \div 10\,\%| = 0.4$**, below 1: inelastic**
2. Incomes rose 10 %; all prices unchanged
   income elasticity: $-4\,\% \div 10\,\% = -0.4$**, below 0: an inferior good**

In pairs, one minute; then hands per item.

Slide 3 Warm-up: which elasticity, what number, what word?

A warm-up on Class 7's elasticities: the same 4 % fall, two causes, two different elasticities. When the good's own price moves, it is the price elasticity of demand, reported without its minus sign; below 1, the quantity moves less than the price: inelastic. When incomes move and prices stay put, it is the income elasticity, and its sign is the message: negative means people buy less as they get richer, an inferior good. Drop that sign and margarine would pass for a normal good. Slide 9 uses the price elasticity again.

Warm-up: which curve moves? Rooms for students

Rooms for students in Barcelona

1. A new campus brings 2,000 more students
   demand for rooms shifts right
2. Owners turn rooms into tourist lets
   supply of rooms shifts left
3. Because of 2, rents rise: does demand shift too?
   no: fewer rooms wanted, along demand

In pairs, one minute; then hands per item.

Slide 4 Warm-up: which curve moves? Rooms for students

The second warm-up is Class 5's shift question, on rooms for students. More students means more buyers at every rent, so the demand curve moves; a room let to tourists is no longer offered to students at any rent, so the supply curve moves. The third item is the trap of the term: once supply has moved the rent rises, and students respond by sharing or living further out, which is a move along the demand curve that is already drawn. One event moves one curve; the change in the good's own price is a move along the other.

16 March 2024: Catalonia caps the rent on new contracts

a new tenant's rent → **no higher than the last tenant's**

140 towns with a stressed rental market, Barcelona included

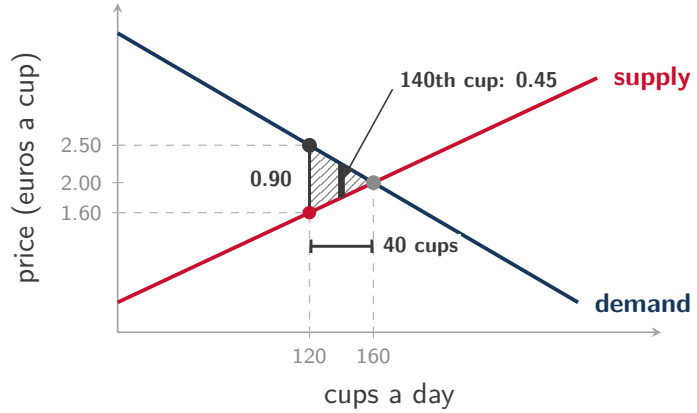
What happens to rents — and to how many flats get rented?

Ley 12/2023 (Spain's housing law); Resolució TER/800/2024
(Diari Oficial de la Generalitat de Catalunya, 14 March 2024), in force 16 March 2024.

Slide 5 16 March 2024: Catalonia caps the rent on new contracts

The hook, from the official gazettes. Spain's 2023 housing law lets a region declare a rental market stressed, and Catalonia was the first region to do so under that law (its 2020 cap, Class 4, had been struck down in 2022): from 16 March 2024, in 140 towns including Barcelona, a new contract may not charge more than the previous tenant paid for the same flat. The slide asks two things: what happens to the rent people sign for, and to how many flats get rented at all. Keep both open while the first half builds the tool; the answer (slide 13) needs Class 8's lost triangle (slide 6) and a cap drawn on Class 7's Gràcia street (slide 11).

Class 8's lost 18 euros: half the tax times the cups not made

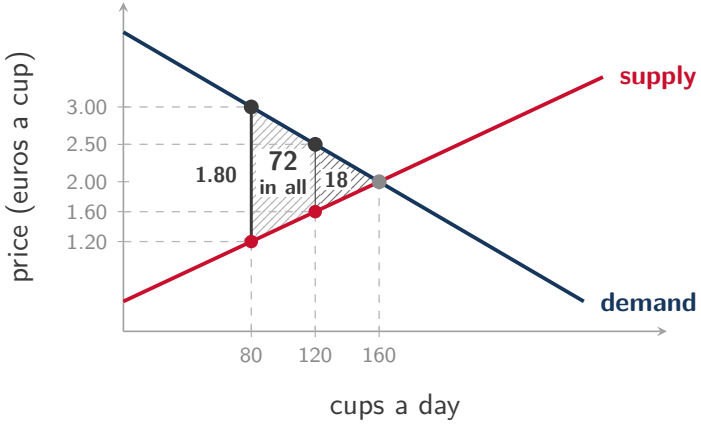


Deadweight loss = ½ × tax × cups lost = ½ × 0.90 × 40 = 18 euros a day.

Slide 6 Class 8's lost 18 euros: half the tax times the cups not made

Monday's picture, with the 18 explained. Take any cup between 120 and 160: someone would pay more for it than a bar needs to make it, so the deal was worth making; with the tax the two sides together cannot cover cost plus tax, the cup is not made, and its surplus goes to nobody. The lost deals form a triangle whose height is the tax and whose base is the cups no longer made, and half of base times height gives the number. The state's revenue is not part of the loss: that money buys something; the deadweight loss is only what disappears. A subsidy has a triangle of the same kind on the other side of 160, where the extra cups cost more than they are worth.

Twice the tax, four times the loss

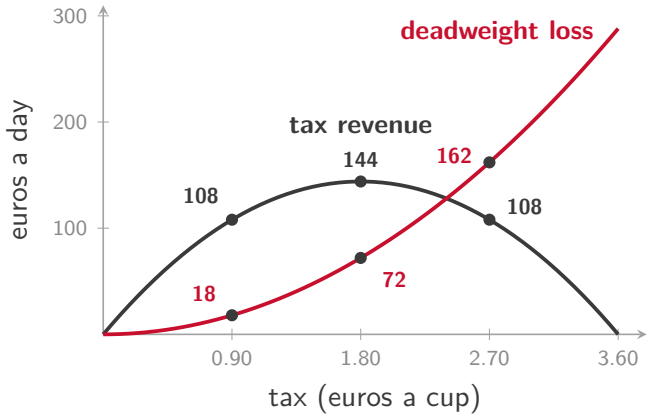


Twice the tax: twice the cups lost, twice the height — four times the loss.

Slide 7 Twice the tax, four times the loss

Same market, the tax doubled. Both sides of the triangle double at once: the wedge is twice as tall, and twice as many cups fall out, because each extra bit of tax stops another band of cups. Half of twice the base times twice the height is four times the area, so the loss grows with the square of the tax: triple it and the loss is nine times. That is why spreading revenue over several small taxes loses less than one big one, which the next slide takes further.

Raise the tax and revenue rises, then falls — the loss keeps growing

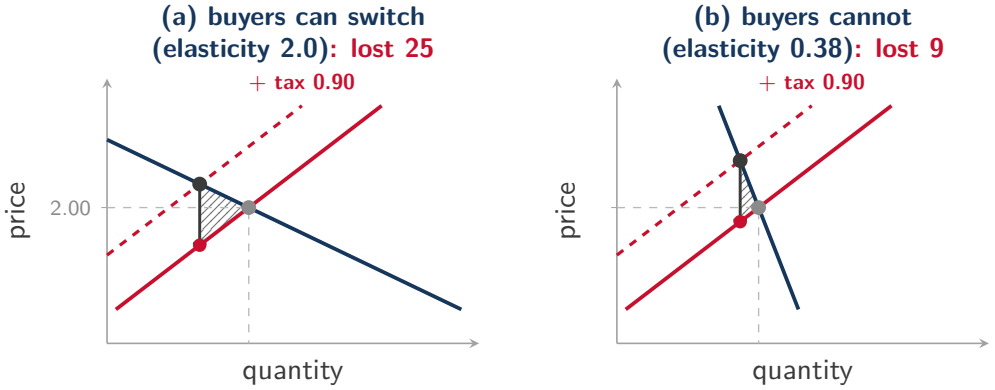


Beyond 1.80 a cup, a higher tax raises less money and loses more.

Slide 8 Raise the tax and revenue rises, then falls — the loss keeps growing

The state's side of the same story. Revenue is the tax times the cups still sold, and the two pull against each other: at first the higher tax wins, after 1.80 the shrinking market wins, and at 3.60 no cup is sold and nothing is collected while the whole of Monday's 288 is gone. The two curves read together are the point: past the top, a higher tax brings in less and loses more. The hump-shaped revenue curve is known as the Laffer curve; the loss curve has no hump, it only rises.

The same 0.90 tax loses more where buyers can walk away



The more elastic the market, the more units a tax stops — and the bigger the loss.

Slide 9 The same 0.90 tax loses more where buyers can walk away

Monday's two taxed markets, now read for what is lost rather than who pays. The supply curve and the tax are identical in both panels; what differs is how many cups the tax stops, and that depends on how easily buyers can walk away. Where they can, many cups drop out and the triangle is wide; where they cannot, few drop out, the triangle is thin, and the state also collects more because the base it taxes stays large. This is one reason tobacco and fuel carry heavy taxes; whether that is fair is a separate question. The same holds on the sellers' side: the more easily they can leave, the bigger the loss.

In pairs, three minutes: what a tax on menús loses

Same street, same menús

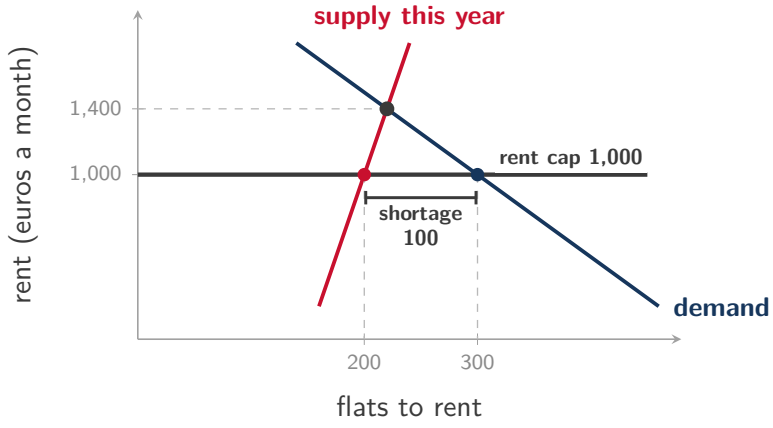
menús a day	the last one is worth	the last one costs
30	15.00	10.00
45	13.50	11.00
60	12.00	12.00

1. Tax 2.50: 45 menús. The lost surplus?
   ½ × 2.50 × 15 = 18.75 a day
2. Tax 5.00: 30 menús. The lost surplus?
   ½ × 5.00 × 30 = 75 a day: four times
3. Revenue at each tax?
   112.50, then 150: up a third

Slide 10 In pairs, three minutes: what a tax on menús loses

Exercise 1 reuses Monday's menú table, so the only new step is the triangle. With a tax the market stops at the row where worth minus cost equals the tax; the base of the triangle is the menús no longer sold, counted from the no-tax 60, not the menús still sold, and the height is the tax. Doubling the tax doubles both and gives four times the loss, slide 7's rule on new numbers. Item 3 is slide 8's point: the state collects only on the menús still sold, so its revenue grows by much less than the loss does.

A rent cap of 1,000 on Gràcia's street: 300 flats wanted, 200 offered



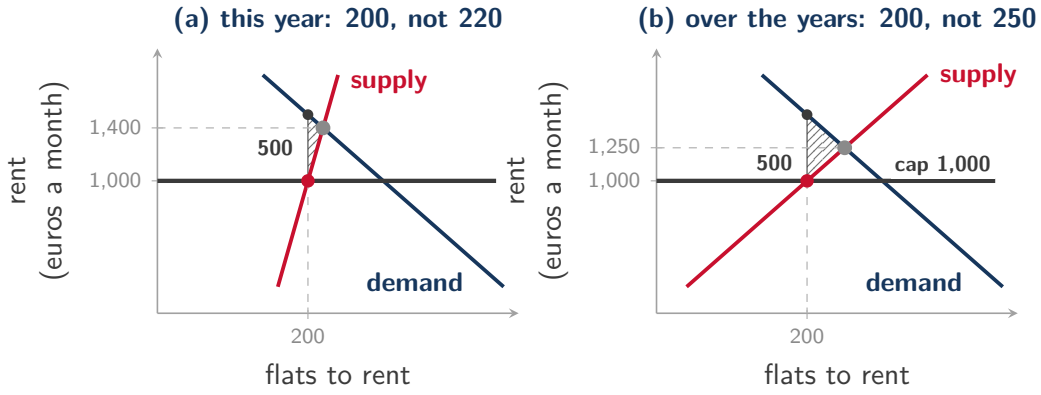
Below the market rent, a cap leaves 100 households searching.

Own numbers (Class 7's Gràcia street); not data.

Slide 11 A rent cap of 1,000 on Gràcia's street: 300 flats wanted, 200 offered

Class 7's Gràcia street, with the government setting the rent. A legal maximum is a price ceiling, and it only matters when it sits below the market rent; a cap above 1,400 would change nothing. At 1,000 owners offer what their supply curve says at that rent and households want what the demand curve says, and the gap between the two is the shortage: not fewer flats in existence, but more wanted than offered at the legal rent. The rent can no longer decide who gets a flat, so something else does: queues, contacts, money under the table, flats let by the season to escape the rule. Those who get a flat pay less; those still searching cannot rent at any price, and whether that trade is worth making is a normative question; what it costs in lost surplus is the next slide.

Over the years the cap costs 50 flats, not 20

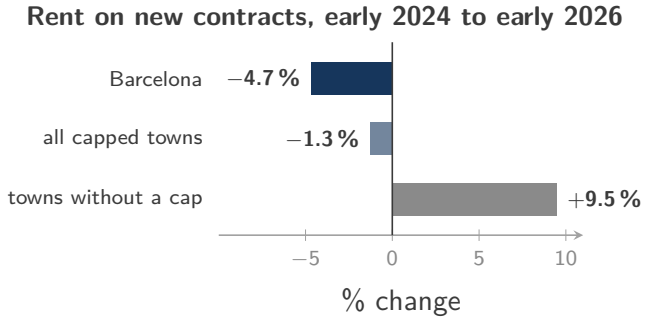


Lost surplus: ½ × 500 × 20 = 5,000 a month this year;
½ × 500 × 50 = 12,500 over the years.

Slide 12 Over the years the cap costs 50 flats, not 20

Why the cap's cost grows. This year the stock of flats is almost fixed, so the cap stops only the 20 flats that the higher rent would have coaxed out; over the years owners build, convert and choose whether to let, supply flattens, and the same cap stops 50. The lost triangle is slide 6's: the 200th flat is worth more to the household that would take it than it costs its owner to offer, and the gap closes only at the market quantity. Demand also loosens with time, so the shortage widens too. The dispute among economists is not the direction but the size: how much supply responds, and how fast, is a positive question.

Catalonia's cap: rents fell — and at first, so did new contracts



−17.2 %
new contracts in capped towns, Apr–Jun 2024 vs Apr–Jun 2023

As the model says: lower rents for those who sign, fewer contracts signed.

Generalitat de Catalunya, from Incasòl's register of rental deposits: press notes of 8 October 2024 (April–June 2024) and 13 July 2026 (first quarter of 2026, against the first quarter of 2024).

Slide 13 Catalonia's cap: rents fell — and at first, so did new contracts

Slide 5's answer, from the one source that sees every contract: every rental deposit in Catalonia must be lodged with the public housing agency, so its register records each new contract and its rent. Over the two years after the cap, rents on new contracts fell in Barcelona and, more mildly, across all the capped towns taken together (Barcelona included), while they rose where there was no cap; and in the first full quarter under the cap, April to June 2024, the capped towns signed about a sixth fewer new contracts than a year earlier. Both are slide 11's predictions: lower rents for those who sign, fewer who get to sign.

Two cautions: fewer contracts is partly tenants moving less once their rent is capped, not only fewer flats; and by early 2026 the number of new contracts was rising again, which the government reads as success while critics point to flats moving to seasonal lets. A raw comparison of capped and uncapped towns is not a study: it says which way, not how much, and how much is where the argument is.

Practice at the board: a cap on allotment rents

Allotment plots (small vegetable gardens) in a city

Rent P in euros a year · gardeners want $Q = 300 - 2P$ plots
Landowners offer $Q = 4P - 120$ · the council caps the rent at 60

1. No cap: the rent and the number of plots?
   70 euros a year, 160 plots
2. With the cap: plots rented, plots wanted?
   120 rented, 180 wanted: a shortage of 60
3. Deadweight loss a year?
   the 120th plot is worth 90: ½ × (90 − 60) × (160 − 120) = 600 euros

Slide 14 Practice at the board: a cap on allotment rents

Slides 11 and 12 in equations. Without the cap, set plots wanted equal to plots offered and solve; the cap matters only because it sits below that rent. At the cap the landowners' equation gives the plots rented, the gardeners' equation the plots wanted; the gap is the shortage. Slide 6's deadweight-loss triangle: its base is the rentals the cap stops, not the shortage; its height is what the last plot still rented is worth to a gardener, the rent at which gardeners want exactly that many, minus the cap.

Break

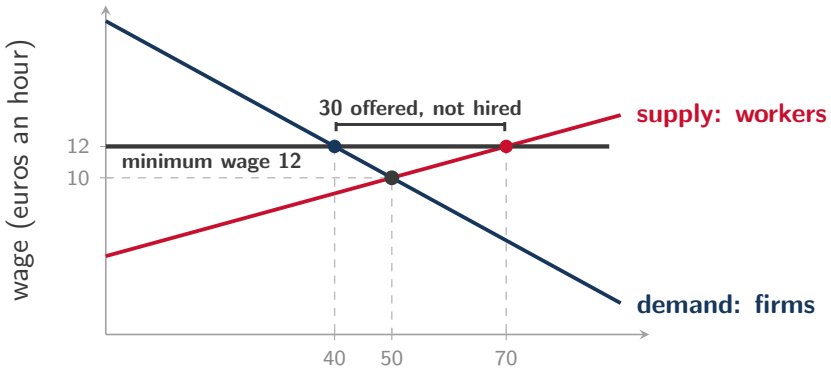
Back at 11:00

Problem set 1: on the front desk as you go out

Slide 15 Break

Ten minutes; Problem set 1 is handed in on the way out. After the break: a legal minimum instead of a maximum, then the border opens.

A minimum wage is a price floor: workers offer 70, firms hire 40

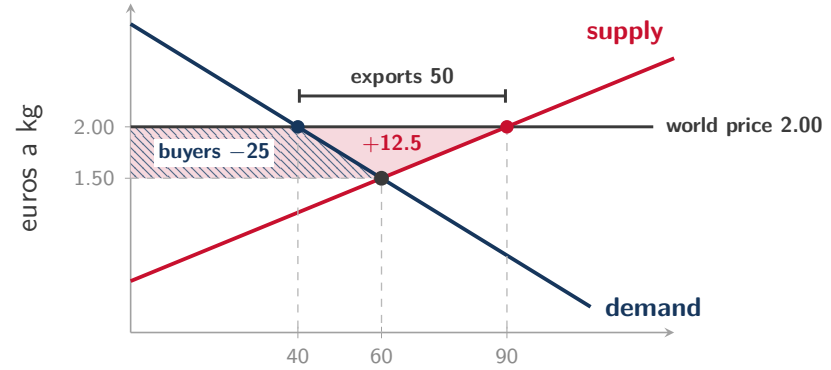


A floor above the market price: sellers offer more than buyers take — here, more hours than firms hire.

Slide 16 A minimum wage is a price floor: workers offer 70, firms hire 40

The mirror of the cap: a price floor, a legal minimum, and the best-known one is the minimum wage. The roles flip: in a labour market the workers are the sellers and the firms the buyers. A floor above the market wage lets those who keep their hours earn more, but firms want fewer hours at the higher wage while workers offer more, and the hours offered but not hired are a surplus, which in a labour market is called unemployment. A floor below the market wage would change nothing, just as a cap above the market rent did nothing. How many hours firms really cut is the disputed number: this picture takes a fifth; real studies find a wide range, smaller for modest rises.

World price above home price: exports; sellers gain, buyers lose less



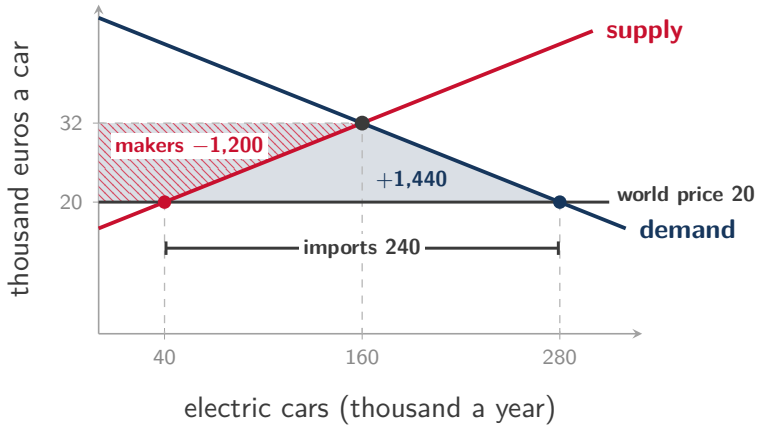
Sellers gain 37.5, buyers lose 25: the country gains 12.5 million euros a year.

1 thousand tonnes = 1 million kg, so euros a kg × thousand tonnes = million euros.

Slide 17 World price above home price: exports; sellers gain, buyers lose less

The border opens, for a country too small to move the world price. If oranges fetch more abroad than at home, no grower sells at home for less, so the home price rises to the world price; at that price home buyers take less, growers sell more, and the difference leaves the country. A world price above the home price is the sign of Class 3's comparative advantage: the country makes oranges at a lower opportunity cost than the rest of the world. Growers gain the whole band between the two prices up to their new quantity; buyers lose the part of that band up to theirs, which simply moves from them to the growers; what is left, the triangle, is a gain nobody had before. The winners could compensate the losers and still be ahead; whether they do is politics, so "exporting makes everyone at home better off" is false.

World price below home price: imports; buyers gain, makers lose less



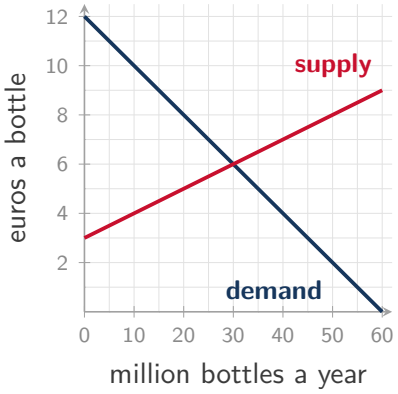
Buyers gain 2,640, home makers lose 1,200: the country gains 1,440 million euros a year.

1 thousand cars × 1 thousand euros a car = 1 million euros.

Slide 18 World price below home price: imports; buyers gain, makers lose less

The other direction. When the world price is below the home price, home makers cannot charge more than an import costs, so the home price falls to the world price; buyers take more, home makers sell less, and imports fill the gap. Now the rest of the world has the comparative advantage. Buyers gain the band between the two prices up to their new quantity, makers lose the part of it up to theirs, and again the difference is a triangle that is pure gain for the country. The rule is the same as for the exporter: the country gains, and the side that now faces competition from abroad loses; "imports make a country poorer" confuses one side's loss with the total.

In pairs, three minutes: cava at home and abroad



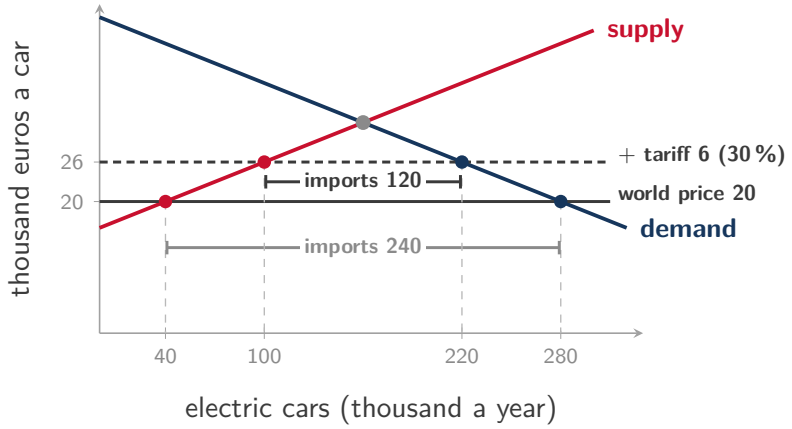
Cava in a small country

1. No trade: price and bottles?
   6 euros, 30 million bottles
2. World price 8: export or import? How many?
   export 30 million: 50 made, 20 drunk
3. At 8, who loses?
   home buyers: pay 8, drink 20 million
4. World price 4 instead?
   import 30 million: 40 drunk, 10 made

Slide 19 In pairs, three minutes: cava at home and abroad

Exercise 2 is the two previous slides on one graph, with a world price that can land on either side of the home price. With a world price above the no-trade price, read the quantity made from the supply curve and the quantity drunk at home from demand; the gap between them, not either quantity, is exported, and home drinkers are the side that loses, because they pay more and drink less. With a world price below it the same reading gives imports, and the roles swap. Either way the country as a whole gains a triangle, half of the price gap times the bottles traded, and it is the same size in both cases here.

A tariff of 6 on imported cars: home price 26, imports halve

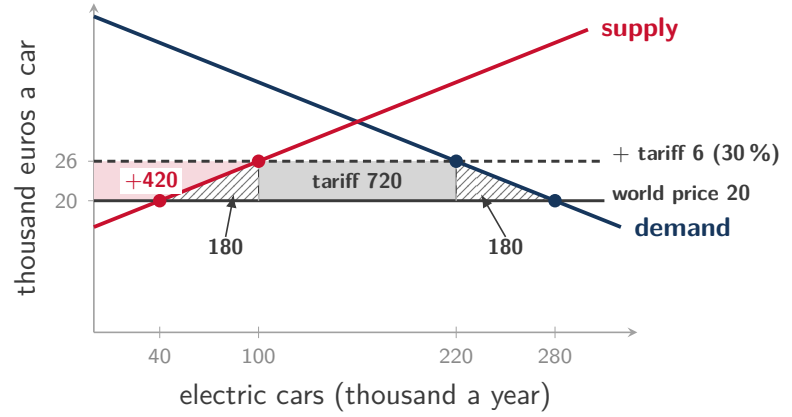


Home price 26: home makers 40 → 100, buyers 280 → 220, imports 240 → 120.

Slide 20 A tariff of 6 on imported cars: home price 26, imports halve

A tariff is a tax on goods made abroad, and in a small country it works like a price rise at the border: an import now costs the world price plus the tariff, home makers can charge the same, and there is one home price for every car wherever it was made. At that higher price home makers move up their supply curve and sell more, buyers move up demand and buy less, so imports shrink from both ends and the market moves back toward where it was without trade. The trap is to think only imported cars get dearer: home-made ones cost the same, because buyers would otherwise switch.

The tariff's bill: 360 million a year that nobody gets



Buyers lose 1,500 = makers 420 + state 720 + 360 that nobody gets (million euros a year).

Slide 21 The tariff's bill: 360 million a year that nobody gets

The tariff's accounts: buyers pay more on every car they still buy and lose the cars they no longer buy, so their loss is the whole band between the two prices up to the old quantity. Part of that band goes to home makers as a higher price on their cars, part goes to the state as tariff on the cars still imported, and two triangles go to nobody: on the left, cars now made at home that cost more than an import would have; on the right, cars that buyers valued above the world price and no longer get. Each triangle is slide 6's formula again, half the tariff times the cars moved. A quota letting in the same number of cars would do the same, except that the holders of the import licences, not the state, would keep the rectangle. In a small country the world price does not move, so the tariff is paid by home buyers, not by the foreign makers.

October 2024: the EU adds duties on electric cars made in China

EU's ordinary duty on cars (China included)	**10 %**
Plus, electric cars made in China:	
BYD	17.0 %
Geely	18.8 %
SAIC	35.3 %
other cooperating makers	20.7 %

February 2026: one China-made model exempted from the extra duty, for a minimum import price and a cap on numbers.

European Commission, IP/24/5589 (29 October 2024) and QANDA/24/3232 (12 June 2024); price undertaking accepted 10 February 2026.

Slide 22 October 2024: the EU adds duties on electric cars made in China

A real tariff, with the rates from the European Commission's own announcement. Since October 2024 battery-electric cars made in China pay the EU's ordinary 10 per cent duty on cars plus a company-specific extra, for five years ("cooperating makers" are those that took part in the Commission's investigation; makers that did not cooperate pay the top rate, 35.3 per cent); the stated reason is that Chinese subsidies threaten European carmakers. The ordinary duty and the extra together are not far from slide 20's 30 per cent, so the same picture applies: a higher price for European buyers, more room for European makers, revenue, and two lost triangles. The February 2026 exception is today's other tools in disguise: one model is let in without the extra duty in exchange for a minimum import price and a limit on numbers, a price floor and a quota.

Five arguments for limiting trade — and the catch

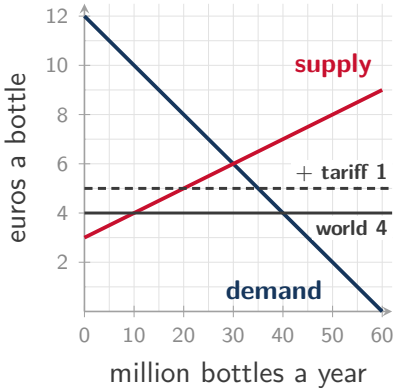
argument	**the catch**
jobs	export industries add jobs
national security	easily stretched to any industry
infant industry	hard to pick future winners
unfair competition	home buyers gain from cheap imports
bargaining chip	threats may have to be carried out

Each protects someone; home buyers pay, two triangles are lost.

Slide 23 Five arguments for limiting trade — and the catch

The reading's five arguments, each with a grain of truth and a catch. Jobs do go, but others grow in the industries with the comparative advantage, and slide 17 showed the winners could pay the losers and still be ahead; national security is fair for a few goods and claimed by every industry; protecting an infant industry needs someone to pick future winners, and politics picks badly; against subsidised imports the home buyers still gain more than home makers lose, so acting anyway is a judgement, Class 4's normative side; and a threat used as a bargaining chip may have to be carried out. Whichever argument wins, the mechanics do not change: home buyers pay, and two triangles are lost.

In pairs, three minutes: a tariff of 1 euro on imported cava



Cava: world price 4, tariff 1

1. Home price and imports?
   5 euros; imports 15 million (35 − 20)
2. Tariff revenue?
   revenue 15 million euros
3. Surplus lost?
   ½ × 1 × 10 + ½ × 1 × 5 = 7.5 million
4. Who gains, and how much?
   home makers +15, the state +15

Slide 24 In pairs, three minutes: a tariff of 1 euro on imported cava

Exercise 3 is slide 21 on the cava market, with the world price below the home price. The tariff lifts the home price by its full amount, since the world price does not move; read the home quantities off the two curves at that price, and imports are the gap between them, smaller than before from both ends. Revenue is the tariff times the bottles still imported, not the old imports. The two lost triangles are not equal here: supply is flatter than demand, so the tariff pulls more extra home bottles into production than it stops drinkers from buying. The makers' gain and the state's revenue come out of the buyers' pocket, and so do the triangles; the revenue is not part of what is lost.

Practice at the board: a tax on driving lessons

Driving lessons in a town, a day; P in euros a lesson

Learners want $Q = 160 - 3P$ lessons · driving schools offer $Q = P$
The town charges the schools 4 euros a lesson

1. No tax: price and lessons?
   40 euros, 40 lessons
2. With the tax: lessons, price learners pay, price schools keep?
   37 lessons; learners pay 41, schools keep 37
3. Tax revenue and deadweight loss?
   revenue 4 × 37 = 148; loss ½ × 4 × 3 = 6 (euros a day)
4. The 4 euros: how much more do learners pay, how much less do schools keep?
   learners 1 more, schools 3 less:
   per euro, learners drop 3 lessons, schools only 1

Slide 25 Practice at the board: a tax on driving lessons

Class 8's tax in today's equations. Learners pay what schools keep plus 4: write the lessons learners want in terms of the price schools keep, set them equal to the lessons offered, and solve. Revenue counts only the lessons still given; the deadweight loss is half the tax times the lessons no longer given. Measure from the no-tax price how much more learners pay and how much less schools keep: the side whose lessons move less per euro, here the schools, carries the larger part, whoever hands the money over.

Practice at the board: dearer pancakes (optional)

A pancake stand raises its price

Price 4 → 6 euros · pancakes sold a day 300 → 180

1. Price elasticity of demand, midpoint method (as a positive number)?
   (120 ÷ 240) ÷ (2 ÷ 5) = 50 % ÷ 40 % = 1.25: elastic
2. Revenue a day: up or down?
   down: 4 × 300 = 1,200 → 6 × 180 = 1,080 euros

Slide 26 Practice at the board: dearer pancakes (optional)

Class 7's elasticity between two points, optional. Take the fall in sales as a percentage of the average quantity, the price rise as a percentage of the average price, and divide the first by the second; measured from the old price and quantity, it lands below one and wrongly predicts higher revenue; only the midpoint gives one answer. Above one, sales fell by a larger percentage than the price rose, so the stand takes in less: price times pancakes at each price confirms it, Class 7's rule for elastic demand.

A tax, a cap or a tariff shrinks a market; twice the wedge, four times the loss.

A binding cap: cheaper for some, a shortage for others.

Open borders: the country gains, some lose; a tariff takes part of it back.
Monday: when a market hurts people outside it.

Slide 27 In one line

The class in four lines, to read slowly: the first is slides 6 to 10 and 21, the second slides 11 to 13, the third slides 17 to 21, and the fourth names Monday.

Before Monday 26 October

- Monday: market failures, externalities and public goods
- Read Mankiw & Taylor: Ch. 9, pp. 204–216; Ch. 8, pp. 189–197 (pp. 197–199 optional)
- Problem set 2 (Block 3): handed out Monday; due Monday 2 November
- Practise in Sextant: the units the course page lists



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Slide 28 Before Monday 26 October

Monday 26 October: what happens when a trade affects people outside it, and goods nobody can be kept from using: market failures, externalities and public goods. Reading: Mankiw & Taylor chapter 9, pages 204 to 216, and chapter 8, pages 189 to 197; pages 197 to 199 are optional. Problem set 2, on Block 3, is handed out on Monday and is due on Monday 2 November. Practise in Sextant with the units the course page lists.

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